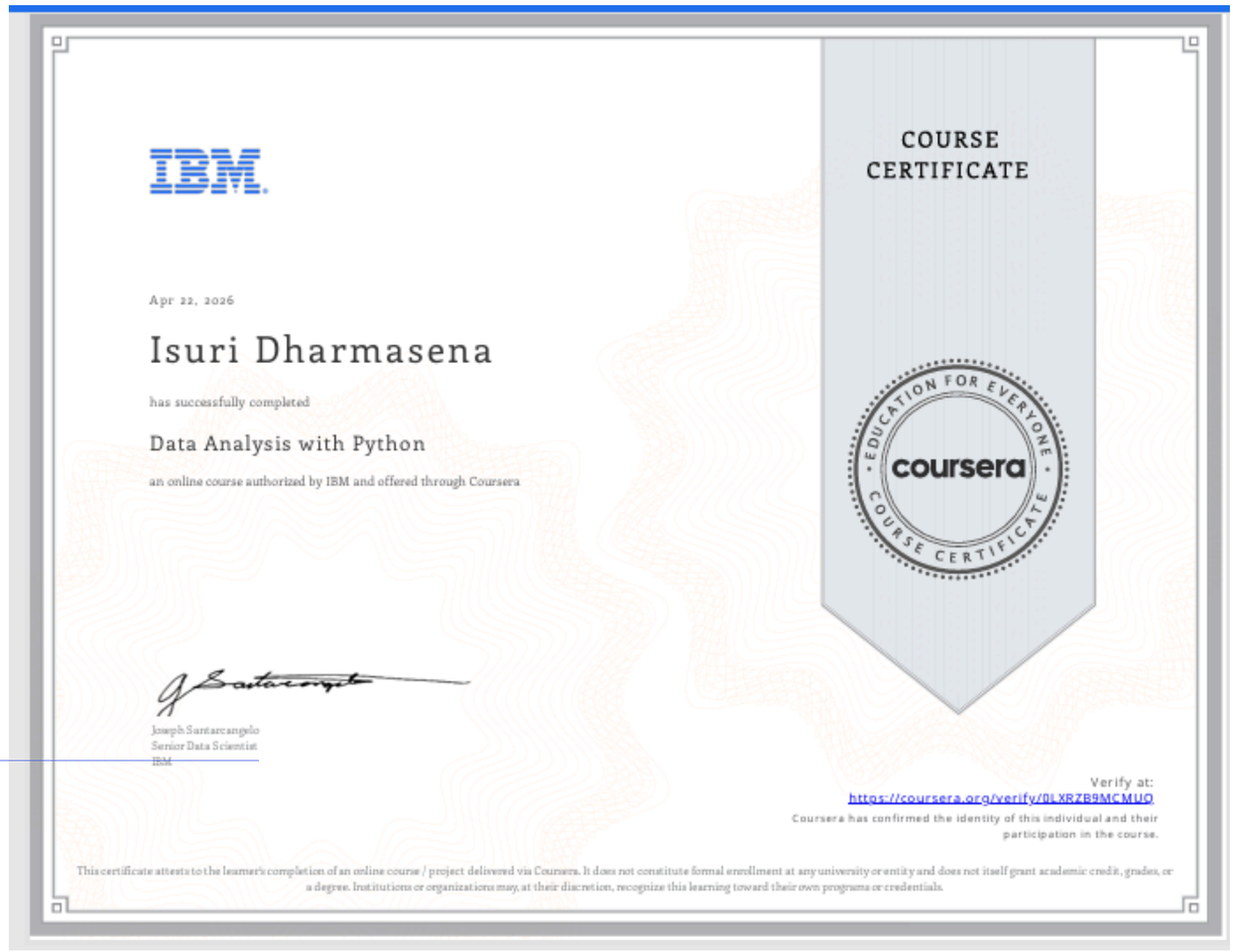
IBM
Project

House Price Prediction with Python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COURSE CERTIFICATE

Apr 22, 2026

Isuri Dharmasena

has successfully completed

Data Analysis with Python

an online course authorized by IBM and offered through Coursera

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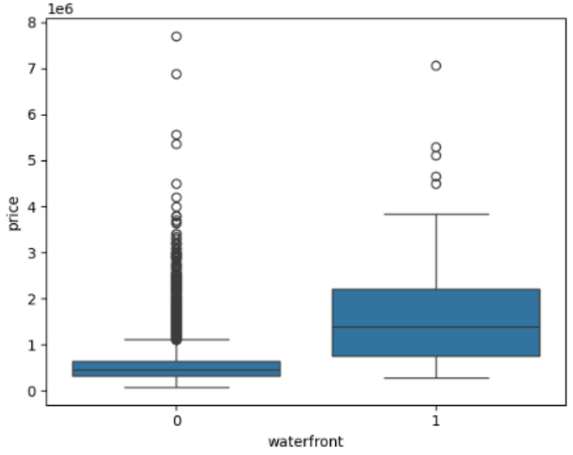
PROJECT TYPE
IBM Data Analysis with Python

FOCUS AREA
King County Housing Data

Data Cleaning • Exploratory Data Analysis • Regression • Ridge Regression • Polynomial Modeling

```
[14]: sns.boxplot(x="waterfront", y="price", data=df)
plt.show()
```



Key Market Insight

LOCATION AND PREMIUM FEATURES STRONGLY INFLUENCE REAL ESTATE VALUE↗

Waterfront properties showed significantly higher median house prices compared to non-waterfront homes. This analysis of the King County housing dataset reveals that premium location features are among the strongest value drivers in real estate markets. Understanding these relationships enables more accurate price predictions and better investment decisions.

3x

Higher median price for waterfront

Model Development

[17]:
```
X = df[['long']]
Y = df['price']
lm = LinearRegression()
lm.fit(X,Y)
lm.score(X, Y)
```

[17]: 0.0004676943014898516

[18]:
```
X = df[['sqft_living']]
Y = df['price']

lm = LinearRegression()
lm.fit(X, Y)

lm.score(X, Y)
```

[18]: 0.4928532179037931

[21]:
```
Input=[('scale',StandardScaler()),('polynomial', PolynomialFeatures(include_bias=False)),('model',LinearRegression())]
```

[22]:
```
pipe = Pipeline(Input)

X = df[features]
Y = df['price']

pipe.fit(X, Y)
pipe.score(X, Y)
```

[22]: 0.7513467656307783

01

LONGITUDE
$R^2 \approx 0.0004$

Longitude shows almost no correlation with house prices, making it unreliable for predictions alone.

02

SQFT LIVING
$R^2 \approx 0.49$

Square footage of living space accounts for nearly half of price variation, proving its predictive power.

03

WEAK VS STRONG PREDICTORS

Weak predictors add noise to models. Strong predictors drive accuracy and meaningful insights.

04

DATA-DRIVEN SELECTION

Smart feature selection separates effective models from unreliable ones in real-world applications.

Feature Strength Matters

Not all variables predict price equally — feature selection is critical for building accurate predictive models.

Model Performance Improved

FEATURE ENGINEERING + SCALING + BETTER MODELING DRAMATICALLY IMPROVE PERFORMANCE

01

LONGITUDE MODEL

$R^2 \approx 0.0004$
Near zero predictive power

02

SQFT LIVING MODEL

$R^2 \approx 0.49$
Moderate predictive power

03

PIPELINE MODEL

$R^2 \approx 0.75$
Strong predictive power

04

KEY TAKEAWAY

Better features + proper scaling = dramatically improved predictions

Final Refined Model

$R^2 = 0.700$

POLYNOMIAL
RIDGE REGRESSION

RIDGE REGRESSION

R^2 = 0.648 baseline performance with regularized linear model.

POLYNOMIAL TRANSFORM

Added polynomial features to capture nonlinear price relationships.

FINAL PERFORMANCE

R^2 = 0.700 achieved through polynomial ridge regression modeling.

MODEL COMPARISON

01

Ridge Regression provided a solid foundation with R^2 = 0.648, effectively handling multicollinearity in housing features while preventing overfitting through L2 regularization.

POLYNOMIAL ENHANCEMENT

02

By applying polynomial feature transformation, the model captured complex nonlinear relationships between square footage, bedrooms, and price that linear models missed.

KEY INSIGHT

03

Polynomial transformation improved predictive accuracy by 8% — demonstrating that housing prices follow nonlinear patterns requiring advanced feature engineering.

Skills Strengthened

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